

L8. Qubit Realizations, Control and Readout

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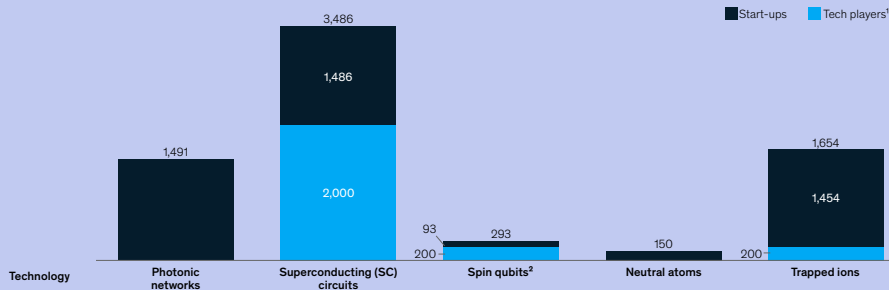
Outline

- Overview of Qubit Technologies
- Superconducting Qubits
- Quantizing the EM Field: Photons
- Coupling Between Atom and EM Field
- Two-Qubit Quantum Gates

11.1 Overview of Qubit Technologies

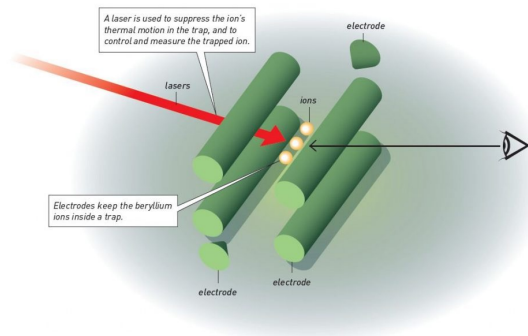
In hardware, tech incumbents invest most in superconducting circuits, while start-ups invest more in ions and photonic networks.

Investment, \$ million



Some Physical Realization of Qubits

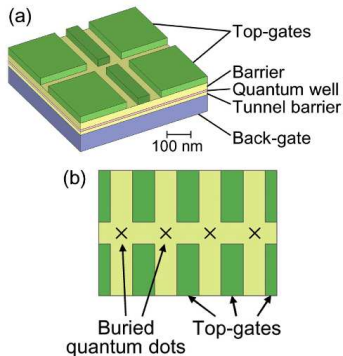
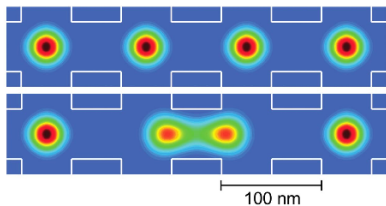
- **Trapped ion qubits:** Use electric and magnetic fields to confine ions in a small region of space. The qubit states are typically encoded in the internal energy levels of the ion.



- Advantages include:

- Long Coherence Times: Ions can be isolated from environmental noise by confining them in a vacuum and using laser cooling techniques to reduce their motion.
- High-Fidelity Operations: Quantum gates and measurements can be performed with very high accuracy using laser pulses.
- Scalability: By creating arrays of ion traps and developing methods for shuttling ions between traps.
- Entanglement and Connectivity: Can be entangled over relatively long distances within the trap array, allowing for complex quantum operations and error correction schemes.

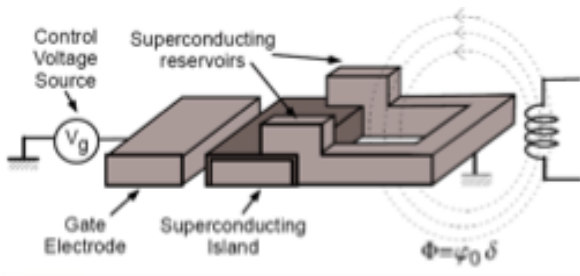
- **Quantum dot qubits:** Use a nanoscale semiconductor structure that confines electrons to store and manipulate quantum information. It acts like an artificial atom, with discrete energy levels similar to those in atomic systems.



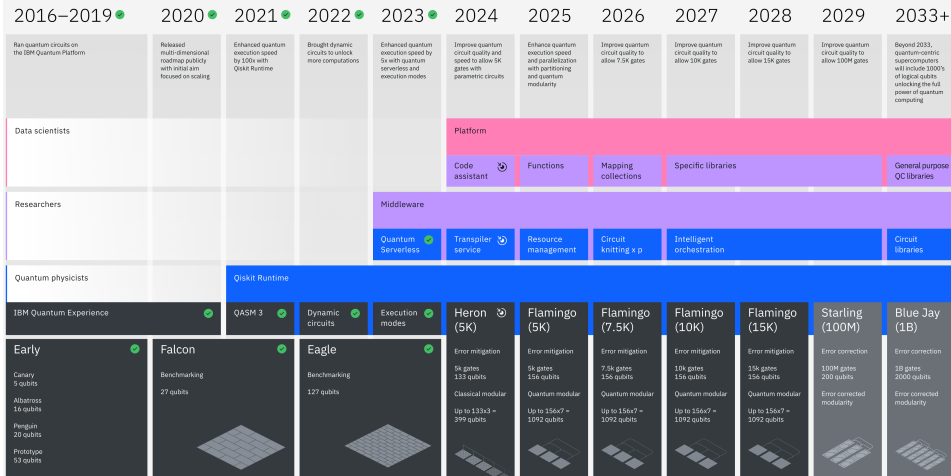
- Advantages include:

- Scalability: Quantum dot qubits can be fabricated using existing semiconductor manufacturing techniques, such as CMOS technology.
- Integration: They can be integrated with classical electronics, which is beneficial for control and readout operations.
- Long Coherence Times: Spin qubits in particular can have relatively long coherence times.

- **Superconducting qubits:** Leading technology in quantum computing, used by Google, IBM, and other companies. Superconducting qubits are artificial atoms built from superconducting circuits. The qubit states are usually defined by the energy levels of the superconducting circuit, which can be manipulated using microwave pulses.



- Advantages include:
 - Long Coherence Times: Exceeding hundreds of microseconds, with some recent demonstrations approaching milliseconds.
 - High-Fidelity Operations: Very accurate quantum gates.
 - Scalability: The fabrication process compatible with existing semiconductor manufacturing techniques.
 - Integration with Classical Electronics: Can be easily integrated with classical control electronics.

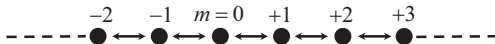


11.2 Superconducting Qubits

- The ground state of a superconductor is a mesoscopic quantum state, characterized by the number of superconducting electron pair, known as the Cooper pairs.



Superconductor Oxide Superconductor



- A Josephson junction creates a weak link in superconductivity. The tunneling of a Cooper pair can be thought of as a particle hopping to an adjacent site on a lattice (in the Cooper pair number space).

- The Josephson junction behaves as a nonlinear inductor with an inductive (kinetic) energy

$$E(\varphi) = -E_J \cos \varphi,$$

where φ is the superconducting phase (can be thought of as the momentum in the number space), such that

$$\hat{n} = i(d/d\varphi).$$

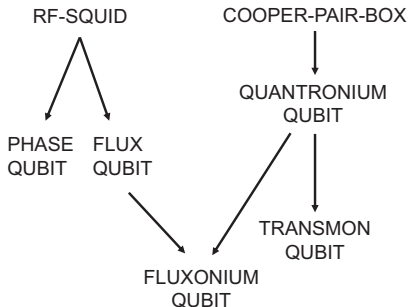
- The I-V characteristics obey the two Josephson relations:

$$I = I_c \sin \varphi, \quad V = \frac{\hbar}{2e} \frac{d\varphi}{dt}.$$

Superconducting Qubits

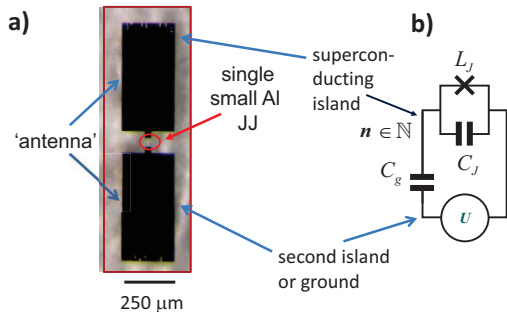
- A Josephson junction, also with its charging energy, realizes an anharmonic oscillator, whose lowest two energy levels can be modeled as a qubit with little dissipation.
- **Transmon qubit:** The junction can be shunted by a large capacitor to reduce its charge noise sensitivity.

Superconducting Qubit Evolutionary Phylogeny



The Cooper Pair Box

- A Cooper pair box qubit has a voltage source biasing the box through a coupling ('gate') capacitor C_g .



- The gate capacitor introduces a continuous offset $n_g = -C_g U / (2e)$, whose fluctuations can be overcome in the transmon qubit design.

- Putting the Josephson energy and the charging energy together, the Hamiltonian is

$$H = 4E_C(\hat{n} - n_g)^2 - E_J \cos \varphi,$$

where $E_C = e^2/(2C)$, and C is the total capacitance between the two superconducting islands.

- For small amplitude oscillations, we make the approximation

$$\cos \varphi \approx 1 - \frac{1}{2}\varphi^2.$$

This is valid in the quantum case in the limit $E_J \gg E_C$, where the zero-point fluctuations in the phase are small.

- Up to an irrelevant constant in the energy (and ignoring the offset charge for the moment) we obtain

$$H \approx 4E_C \hat{n}^2 + \frac{1}{2} E_J \hat{\varphi}^2,$$

where we recall that

$$\hat{n} = i(d/d\varphi).$$

- Drawing an analogy to a harmonic oscillator (see next page), we find the resonant frequency of the CPB

$$\Omega_J = \frac{1}{\hbar} \sqrt{8E_J E_C}.$$

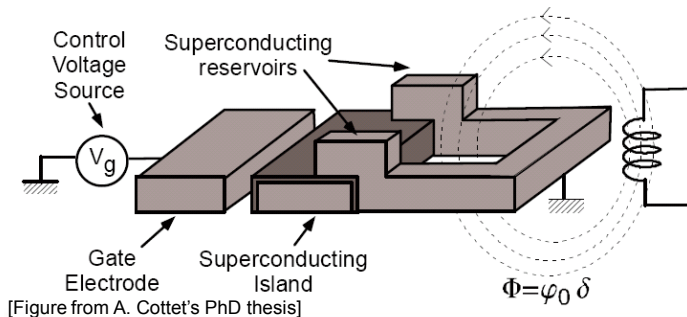
	Harmonic Oscillator	Cooper Pair Box
Position	x	$\hat{n}$
Momentum	p	$\hbar\hat{\phi}$
Kinetic Energy	$p^2/(2m)$	$(E_J/2)\hat{\phi}^2$
Potential Energy	$kx^2/2$	$4E_C\hat{n}^2$
Resonant ω	$\sqrt{k/m}$	$\sqrt{8E_JE_C}/\hbar$

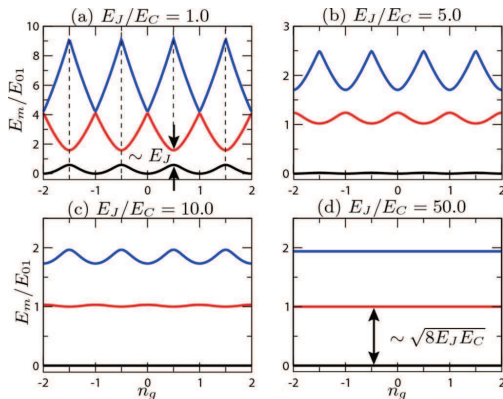
- The resonant frequency is also known as the Josephson plasma frequency

$$\Omega_J = \frac{1}{\sqrt{L_J(\varphi = 0)C}}, \quad L_J = \left(\frac{\hbar}{2e}\right)^2 \frac{1}{E_J \cos \varphi}.$$

- The approximation we made replaces the Josephson junction by a linear inductance $L = L_J(\varphi = 0)$.
- The non-linear inductance of the junction, or the $O(\varphi^4)$ terms, will make the energy levels of the Cooper pair box anharmonic.

- Offset charge n_g tuned by gate (translation of $\hat{n}$)
- Josephson energy E_J tuned by flux in split CPB
- Charging energy E_C , usually fixed by geometry





- Effects of increasing $E_J/E_C \longrightarrow$ transmon qubit
 - Good: Flatter energy levels, insensitive to charge fluctuations
 - Bad: Decreasing anharmonicity

Temperature

- The superconducting transition temperature T_c of Al is about 1 K, which can be converted to 21 GHz.
- So the operating temperature must satisfy $T \ll T_c$. This is achieved by working inside a dilution refrigerator with base temperature about 30 mK.
- The transition frequency of a qubit is usually a few GHz, which is much greater than T , so thermal fluctuations are suppressed. Nevertheless, experiments have found a surprisingly large amount of nonequilibrium quasiparticles that lead to qubit energy relaxation $|1\rangle \rightarrow |0\rangle$.

11.3 Quantizing the EM Field in a Cavity

Take-Home Messages

- The EM field in the cavity is quantized: each mode is a harmonic oscillator.
- For a single mode, the electric field operator is

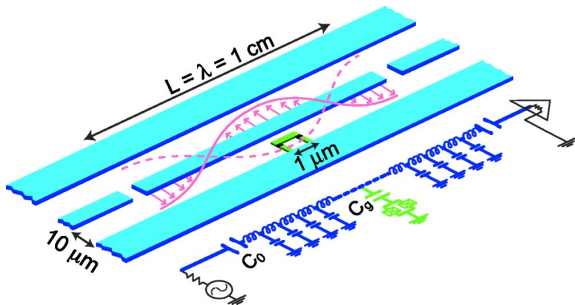
$$\vec{E}_c = E_{c0}(\hat{\epsilon}a + \hat{\epsilon}^*a^+).$$

- A small cavity enhances quantum fluctuations of the electric field ($\sim E_{c0}$).

Coupling a Qubit to the EM Field in a Cavity

- By adjusting the irradiating EM wave frequency, one can either perform a qubit rotation (at the qubit transition frequency—classical Rabi oscillations) or a measurement of the state (near the cavity frequency).
- In the latter, the qubit is like a medium and the qubit and does not exchange energy directly with the cavity. Instead, they are coupled via virtual photons. As a result, the cavity's frequency is no longer a fixed constant; it depends on the qubit state.

- To realize such a cavity in circuit QED, one can use a coplanar waveguide (CPW) resonator formed by superconducting wires.



- One can understand qubit measurement and qubit-qubit interaction in the framework of quantum electrodynamics (QED), which studies atoms and electrons coupled to quantum fluctuations of the electromagnetic (EM) field.
 - Cavity QED engineers those quantum fluctuations by surrounding the atoms with a resonant cavity which supports only discrete modes of the EM field.
 - Circuit QED engineers superconducting qubits as artificial atoms coupled to microwave resonators (typically, with a single mode).

Quantizing the EM Field: Photons

- The Hamiltonian (or energy) of the EM field in a cavity is

$$H = \int_V d^3r \left[\frac{1}{2} \epsilon_0 E^2(\vec{r}, t) + \frac{1}{2\mu_0} B^2(\vec{r}, t) \right].$$

- According to Maxwell's equations in a cavity, we can, roughly speaking, regard E as x and B as p . This leads to the quantization of the EM field in the cavity: Each mode is a harmonic oscillator. This is not unlike the discrete modes of a musical instrument getting quantized.

- The energy of a single mode with frequency ω_c is

$$E_n = (n + 1/2)\hbar\omega_c,$$

which implies that the mode is occupied by n photons (they are bosons!). We label the corresponding eigenstates as $|n\rangle$; this is the number basis.

- In the number basis, we have
 - $\hat{N}|n\rangle \equiv a^\dagger a|n\rangle = n|n\rangle$.
 - $a|n\rangle = \sqrt{n}|n-1\rangle$, where $\langle n|n\rangle = 1$.

- In the picture of quantum field, the electric field operator is

$$\vec{E}_c = E_{c0}(\hat{\epsilon}a + \hat{\epsilon}^*a^+),$$

where a is the photon annihilation operator and $\hat{\epsilon}$ the complex polarization vector.

- In the number basis, we can write

$$a = \sum_n \sqrt{n+1} |n\rangle \langle n+1| = \sum_n \sqrt{n} |n-1\rangle \langle n|,$$
$$a^+ = \sum_n \sqrt{n+1} |n+1\rangle \langle n|.$$

- To calculate matrix elements, we write $\langle n-1|a|n\rangle = \sqrt{n}$ and $\langle n+1|a^+|n\rangle = \sqrt{n+1}$.

11.4 Coupling Between Qubits and Photons

Take-Home Messages

- The dipole interaction of the qubit and EM field can be approximated, in the rotating-wave approximation by the Janyes-Cummings model.
- The Janyes-Cummings model can be solved, in particular, in the dispersive case, which leads to quantum non-demolition readout.
- Similarly, two qubits can be coupled via a cavity, which facilitates the realization of two-qubit gates.

Coupling Between Qubits and Photons

- With the quantized field, we can establish a fully quantum-mechanical model to describe the interaction between qubits and photons.
- The Jaynes-Cummings model has a Hamiltonian

$$H_{JC} = \underbrace{-\frac{1}{2}\hbar\omega_{01}\sigma_z}_{H_{\text{qubit}}} + \underbrace{\hbar\omega_c(a^+a + 1/2)}_{H_{\text{field}}} + \underbrace{\hbar g(a\sigma_+ + a^+\sigma_-)}_{H_{\text{int}}},$$

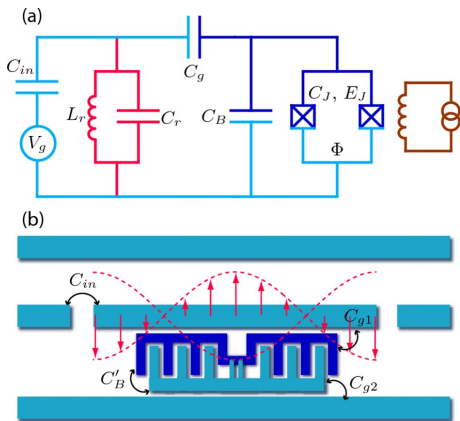
where $\sigma_{\pm} = (\sigma_x \pm i\sigma_y)/2$. Notice that $\sigma_+ |g\rangle = |e\rangle$, $\sigma_- |e\rangle = |g\rangle$, and $\sigma_+ |e\rangle = \sigma_- |g\rangle = 0$.

- The quantum Hamiltonian arises from a coupling between the electric field $\vec{E}$ in a single-mode cavity and the dipole moment $\vec{p}$ of the artificial atom at position $\vec{R}$ given by

$$U = -\vec{p} \cdot \vec{E}(\vec{R}).$$

- If the cavity mode is close in frequency to the qubit transition frequency, we drop the terms that significantly change the total energy, in the so-called **rotating wave approximation**.
- Thereby, we arrive at the Jaynes-Cummings Hamiltonian, which may further include external driving and damping terms.

- The circuit design of the transmon-cavity coupling is discussed in J. Koch et al., Phys. Rev. A 76, 042319 (2007).



Solving the Jaynes-Cummings Hamiltonian

- The Hilbert space is the tensor product of a harmonic oscillator and a qubit, whose basis vectors are

$$|0, g\rangle, |1, g\rangle, |2, g\rangle, \dots,$$

$$|0, e\rangle, |1, e\rangle, |2, e\rangle, \dots,$$

where the first index is the number of photons and the second the qubit state ($g \equiv 0$, $e \equiv 1$).

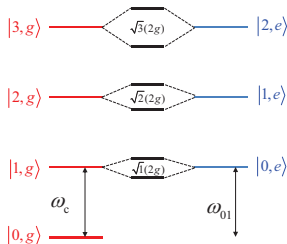
- When the temperature is low and when we are not interested in pumping in many photons, we can truncate the Hilbert space to include just a few basis states.

- Let us consider the case where the detuning between the qubit and cavity frequencies

$$\Delta \equiv \omega_{01} - \omega_c = 0.$$

Photon and qubit excitations hybrid into the so-called **polaritons**.

- In this case, the two states $|n+1, g\rangle$ and $|n, e\rangle$ are degenerate in the absence of interaction.



- The dipole coupling lift the degeneracy by introducing off-diagonal matrix elements (which are real)

$$\langle n+1, g | H_{\text{int}} | n, e \rangle = \langle n, e | H_{\text{int}} | n+1, g \rangle = \sqrt{n+1} \hbar g.$$

- In the truncated 2D subspace, the effective Hamiltonian is

$$\begin{aligned} H' &= \begin{pmatrix} (n+1)\hbar\omega_c & \sqrt{n+1}\hbar g \\ \sqrt{n+1}\hbar g & (n+1)\hbar\omega_c \end{pmatrix} \\ &= (n+1)\hbar\omega_c I_{2 \times 2} + \sqrt{n+1}\hbar g \sigma_x. \end{aligned}$$

- The energy eigenstates are coherent superpositions

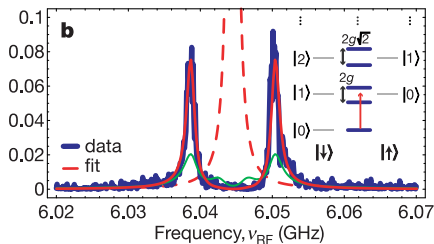
$$|\Psi_{\pm}\rangle = \frac{1}{\sqrt{2}}(|n+1, g\rangle \pm |n, e\rangle),$$

i.e., the bonding–anti-bonding combination of photon and qubit excitation, known as polaritons.

- The corresponding energies are

$$E_{\pm} = (n+1)\hbar\omega_c \pm \sqrt{n+1}\hbar g.$$

- In an experimental breakthrough at Yale, Andreas Wallraff and collaborators have demonstrated for the first time that a solid-state nano-electronic quantum circuit can be coupled coherently to a single microwave photon [Nature 431, 162 (2004)].



- In the so-called **dispersive regime**, where $|\Delta| \gg g$, the truncated Hamiltonian becomes

$$\begin{aligned}
 H' &= \begin{pmatrix} (n+1)\hbar\omega_c & \sqrt{n+1}\hbar g \\ \sqrt{n+1}\hbar g & n\hbar\omega_c + \hbar\omega_{01} \end{pmatrix} \\
 &= \epsilon_n I_{2 \times 2} + \sqrt{n+1}\hbar g \sigma_x - \frac{\hbar\Delta}{2} \sigma_z.
 \end{aligned}$$

where

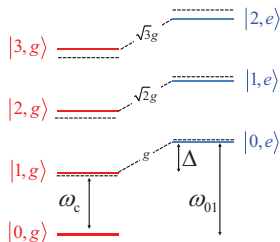
$$\epsilon_n = \left(n + \frac{1}{2}\right) \hbar\omega_c + \frac{1}{2} \hbar\omega_{01}.$$

- One can verify that H' reduces to the degenerate form when $\Delta \equiv \omega_{01} - \omega_c = 0$.

- The coupling introduces an effective potential, up to a constant,

$$V_{\text{dispersive}} = -\hbar \frac{g^2}{\Delta} \left[a^\dagger a + \frac{1}{2} \right] \sigma_z,$$

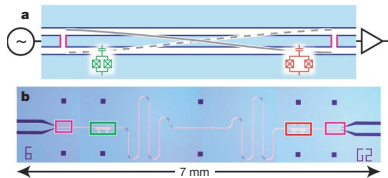
in the limit of $g \rightarrow 0$.



- This is result of a second-order perturbation. It can be interpreted as a shift in the cavity frequency which depends on the state of the qubit, thus can be measured without directly disturbing the qubit state.
- Therefore, the coupling is quantum non-demolition (QND) with respect to both photon number and qubit polarization.

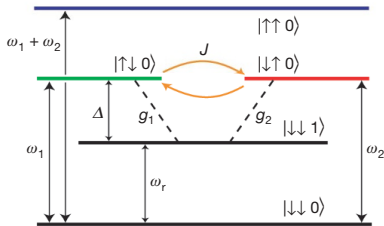
11.5 Two Qubits Coupled via a Cavity

- Two qubits connected to the same resonator can be coupled passively through its virtual excitations.
- The effective Hamiltonian for the resonator and two such qubits can be obtained as



$$H_{12}^{\text{eff}} = \left[\hbar\omega_c \left(a^+ a + \frac{1}{2} \right) \right]_{\text{field}} - \left[\frac{\hbar\omega_{01}}{2} (\sigma_{1z} + \sigma_{2z}) \right]_{\text{qubits}} \\ - \frac{\hbar g^2}{\Delta} \left(a^+ a + \frac{1}{2} \right) (\sigma_{1z} + \sigma_{2z}) - \frac{\hbar g^2}{\Delta} (\sigma_{1+}\sigma_{2-} + \sigma_{1-}\sigma_{2+}).$$

- The qubits should be tuned in resonance with each other, but away from the resonator's frequency.
- In the qubit subspace,



$$U(t) = e^{i\frac{\hbar g^2 t}{\Delta}(a^+ a + \frac{1}{2})}(\sigma_{1z} + \sigma_{2z}) \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos \frac{g^2}{\Delta} t & i \sin \frac{g^2}{\Delta} t & 0 \\ 0 & i \sin \frac{g^2}{\Delta} t & \cos \frac{g^2}{\Delta} t & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

- This does not change the state of the resonator mode. At $t = \pi|\Delta|/(4g^2)$, it corresponds to a $\sqrt{i\text{SWAP}}$ operation.

Summary

- Superconductivity introduces a spectral gap, hence greatly suppresses low-energy excitation and dissipation.
- The Josephson effect leads to superconductivity with a weak link and a nonlinear inductance.
- Therefore, a Josephson junction, also with its charging energy, realizes an anharmonic oscillator, whose lowest two energy levels can be modeled as a qubit with little dissipation.
- The control and readout of qubits can be realized by coupling qubits to quantized EM field, or photons in a cavity.

References

- S. M. Girvin, Circuit QED: Superconducting Qubits Coupled to Microwave Photons, Proceedings of the 2011 Les Houches Summer School on Quantum Machines, Oxford University Press (2014).
- For further reading on transmon qubits, see, e.g., T. E. Roth, R. Ma, W. C. Chew, An Introduction to the Transmon Qubit for Electromagnetic Engineers, arXiv:2106.11352 (2021).